Lab 2: Network Commands for Testing and Troubleshooting

Required Tools

- A computer with command line interface (CLI) access
- Network connection (wired or wireless)

Objective

To familiarize with essential network commands used for testing and troubleshooting network connectivity issues.

Theory

Introduction to Network Troubleshooting

Network troubleshooting is a systematic approach to identifying and resolving connectivity issues in computer networks. Command-line tools provide network administrators with powerful utilities to analyze network behavior, test connectivity, and diagnose problems efficiently.

The OSI Model and Network Commands

Network commands operate at different layers of the OSI model. Commands like ping and tracert work at the Network Layer (Layer 3) for testing connectivity and routing. The arp command operates at the Data Link Layer (Layer 2) for MAC address resolution, while netstat monitors Transport Layer (Layer 4) connections. Application layer tools like nslookup help diagnose DNS issues.

Key Networking Concepts

TCP/IP Protocol: The fundamental communication protocol suite for networks. IP handles addressing and routing, while TCP ensures reliable data delivery.

DNS (Domain Name System): Translates human-readable domain names into IP addresses that computers use for communication.

ARP (Address Resolution Protocol): Maps IP addresses to physical MAC addresses on local networks.

Routing: The process of selecting network paths to send data packets. Routing tables determine the best path for packet forwarding.

Network Troubleshooting Methodology

1. Identify the problem and gather information
2. Test connectivity using diagnostic commands
3. Analyze command outputs to isolate the issue
4. Implement corrective measures
5. Verify that the problem is resolved

Procedure

1. Open the command line interface (CLI) on your computer:
    - **Windows:** Press Win + R, type cmd, and press Enter
    - **Linux/Mac:** Open Terminal application
2. Use the ipconfig or ifconfig command to check your current network configuration
3. Execute each command listed above following the provided syntax
4. Document the output of each command by taking screenshots
5. Analyze the results to understand your network configuration and connectivity

Network Commands: Syntax, Usage and Output:

1. ping

Syntax: ping [hostname or IP address]

Usage: Tests the reachability of a host on an IP network and measures the round-trip time for messages sent from the originating host to a destination computer.

Theory: Sends ICMP Echo Request packets to the target and waits for replies. Reports packet loss and response times to verify basic connectivity.

```
C:\Users\HP>ping www.google.com

Pinging www.google.com [142.250.67.68] with 32 bytes of data:
Reply from 142.250.67.68: bytes=32 time=68ms TTL=114
Reply from 142.250.67.68: bytes=32 time=21ms TTL=114
Reply from 142.250.67.68: bytes=32 time=22ms TTL=114
Reply from 142.250.67.68: bytes=32 time=22ms TTL=114

Ping statistics for 142.250.67.68:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 21ms, Maximum = 68ms, Average = 33ms
```

2. ipconfig

Syntax: ipconfig

Usage: Displays all current TCP/IP network configuration values and refreshes DHCP and DNS settings.

Theory: Shows network adapter information including IP address, subnet mask, default gateway, and DNS servers. First command used when troubleshooting network issues.

```
rosoft Windows [Version 10.0.26100.7462]
 Microsoft Corporation. All rights reserved.

Windows\System32>ipconfig /all

dows IP Configuration

Host Name . . . . . : LAPTOP-D4UFQDGJ
Primary Dns Suffix  . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

eless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 68-54-5A-AD-EF-4F
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

eless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : 6A-54-5A-AD-EF-4E
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

eless LAN adapter Wi-Fi:

Connection-specific DNS Suffix  . :
Description . . . . . : Intel(R) Wi-Fi 6 AX200 160MHz
Physical Address. . . . . : 68-54-5A-AD-EF-4E
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::b25b:16d9:7c03:357%17(Preferred)
IPv4 Address. . . . . : 10.5.5.200(Preferred)
Subnet Mask . . . . . : 255.255.0.0
Lease Obtained. . . . . : Thursday, January 8, 2026 10:27:12 AM
Lease Expires . . . . . : Saturday, January 10, 2026 10:27:12 PM
Default Gateway . . . . . : 10.5.50.1
DHCP Server . . . . . : 10.5.50.1
DHCPv6 IAID . . . . . : 191386714
DHCPv6 Client DUID. . . . . : 00-01-00-01-29-E9-F0-2B-68-54-5A-AD-EF-4E
DNS Servers . . . . . : 202.79.32.4
                                185.228.168.9
NetBIOS over Tcpip. . . . . : Enabled
```

3. tracert

Syntax: tracert [hostname or IP address]

Usage: Determines the route taken by packets to reach a specific host by listing all the intermediate routers.

Theory: Sends packets with increasing TTL values. Each router along the path returns its IP address, mapping the complete route from source to destination.

```
C:\Users\HP>tracert www.google.com

Tracing route to www.google.com [142.250.67.68]
over a maximum of 30 hops:

  1     8 ms     2 ms     2 ms  10.5.50.1
  2    50 ms     4 ms     2 ms  172.25.25.1
  3     7 ms     6 ms     9 ms  1.220.166.202.ether.static.wlink.com.np [202.166.220.1]
  4    15 ms     6 ms    11 ms  be-17.3005-140.40.gwc-dhub-core-01.wlink.com.np [202.79.40.140]
  5     8 ms     7 ms     6 ms  ae-21-170.41.gwmj-lknt-01.wlink.com.np [202.79.41.170]
  6     6 ms     5 ms     7 ms  ae-20-128.41.gwj-btwl-core-01.wlink.com.np [202.79.41.128]
  7    11 ms     8 ms    11 ms  ae52-ipt-bhwa-01.wlink.com.np [72.9.128.67]
  8     *        *        *     Request timed out.
  9    24 ms    19 ms    20 ms  142.250.174.2
 10    21 ms    21 ms    20 ms  142.250.208.225
 11    20 ms    17 ms    18 ms  142.251.54.97
 12    21 ms    20 ms    20 ms  maa05s13-in-f4.1e100.net [142.250.67.68]

Trace complete.
```

4. netstat

Syntax: netstat -a

Usage: Displays active TCP connections, ports on which the computer is listening, Ethernet statistics, and more.

Theory: Provides visibility into network connections, identifies which applications are using network resources, and verifies that services are listening on correct ports.

Common options:

- netstat -a - All connections and listening ports
- netstat -n - Numerical addresses and ports
- netstat -o - Shows process ID (PID)
- netstat -r - Displays routing table

```
C:\Users\HP>netstat -a

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:135            LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    0.0.0.0:445            LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    0.0.0.0:5040           LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    0.0.0.0:5357           LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    0.0.0.0:6646           LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    0.0.0.0:49664          LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    0.0.0.0:49665          LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    0.0.0.0:49666          LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    0.0.0.0:49667          LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    0.0.0.0:49668          LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    0.0.0.0:49669          LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    10.5.5.200:139         LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    10.5.5.200:49351       a2-17-106-145:https    ESTABLISHED
  TCP    10.5.5.200:49352       40.126.16.166:https    ESTABLISHED
  TCP    10.5.5.200:49353       135.234.160.245:https  TIME_WAIT
  TCP    10.5.5.200:49355       52.111.252.1:https     ESTABLISHED
  TCP    10.5.5.200:50901       a23-217-153-205:https  CLOSE_WAIT
  TCP    10.5.5.200:51254       ns5033994:https        TIME_WAIT
  TCP    10.5.5.200:51320       ec2-52-0-252-3:https   ESTABLISHED
  TCP    10.5.5.200:51610       pnmaaa-ax-in-f10:https  ESTABLISHED
  TCP    10.5.5.200:53768       sm-in-f188:5228        ESTABLISHED
  TCP    10.5.5.200:55433       4.153.25.42:https      CLOSE_WAIT
  TCP    10.5.5.200:57904       52.111.252.1:https     ESTABLISHED
  TCP    10.5.5.200:58347       ec2-18-97-36-67:https  ESTABLISHED
  TCP    10.5.5.200:59543       adult-filter-dns:https  ESTABLISHED
  TCP    10.5.5.200:61126       52.98.59.18:https      ESTABLISHED
  TCP    10.5.5.200:62492       103.211.149.154:https  ESTABLISHED
  TCP    10.5.5.200:62494       52.111.252.1:https     ESTABLISHED
  TCP    10.5.5.200:62742       4.213.25.241:https     ESTABLISHED
  TCP    10.5.5.200:63407       249:https              TIME_WAIT
  TCP    10.5.5.200:65041       13.66.226.80:https     CLOSE_WAIT
  TCP    127.0.0.1:45112        LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    127.0.0.1:51316        LAPTOP-D4UFQDGJ:51317  ESTABLISHED
  TCP    127.0.0.1:51317        LAPTOP-D4UFQDGJ:51316  ESTABLISHED
  TCP    127.0.0.1:51318        LAPTOP-D4UFQDGJ:51319  ESTABLISHED
  TCP    127.0.0.1:51319        LAPTOP-D4UFQDGJ:51318  ESTABLISHED
  TCP    127.0.0.1:64150        LAPTOP-D4UFQDGJ:64151  ESTABLISHED
  TCP    127.0.0.1:64151        LAPTOP-D4UFQDGJ:64150  ESTABLISHED
  TCP    127.0.0.1:64152        LAPTOP-D4UFQDGJ:64153  ESTABLISHED
  TCP    127.0.0.1:64153        LAPTOP-D4UFQDGJ:64152  ESTABLISHED
  TCP    [::]:135               LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    [::]:445               LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    [::]:5357              LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    [::]:49664             LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    [::]:49665             LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    [::]:49666             LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    [::]:49667             LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    [::]:49668             LAPTOP-D4UFQDGJ:0      LISTENING
  TCP    [::]:49669             LAPTOP-D4UFQDGJ:0      LISTENING
  UDP    0.0.0.0:123            *:*
  UDP    0.0.0.0:3702           *:*
  UDP    0.0.0.0:3702           *:*
  UDP    0.0.0.0:5050           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5355           *:*
  UDP    0.0.0.0:6646           *:*
  UDP    0.0.0.0:49864          *:*
  UDP    0.0.0.0:52042          *:*
  UDP    10.5.5.200:137         *:*
  UDP    10.5.5.200:138         *:*
  UDP    10.5.5.200:1900        *:*
  UDP    10.5.5.200:2177        *:*
  UDP    10.5.5.200:58193       *:*
  UDP    127.0.0.1:1900         *:*
  UDP    127.0.0.1:49664        127.0.0.1:49664
  UDP    127.0.0.1:58194        *:*
  UDP    [::]:123               *:*
  UDP    [::]:3702              *:*
  UDP    [::]:3702              *:*
  UDP    [::]:5353              *:*
  UDP    [::]:5353              *:*
  UDP    [::]:5355              *:*
  UDP    [::]:49865             *:*
  UDP    [::]:52042             *:*
  UDP    [::1]:1900             *:*
  UDP    [::1]:58192            *:*
  UDP    [fe80::b25b:16d9:7c03:357%17]:1900  *:*
  UDP    [fe80::b25b:16d9:7c03:357%17]:2177  *:*
  UDP    [fe80::b25b:16d9:7c03:357%17]:58191  *:*
```

5. nslookup

Syntax: nslookup [hostname]

Usage: Queries the Domain Name System (DNS) to obtain domain name or IP address mapping information.

Theory: Performs DNS queries to resolve hostnames to IP addresses. Essential for diagnosing DNS-related problems.

```
C:\Users\HP>nslookup www.google.com
Server:  ns.wlink.com.np
Address:  202.79.32.4

Non-authoritative answer:
Name:    www.google.com
Addresses:  2404:6800:4002:806::2004
          142.250.67.68
```

6. arp

Syntax: arp -a

Usage: Displays and modifies the IP-to-Physical (MAC) address translation table used by the Address Resolution Protocol (ARP).

Theory: Shows IP-to-MAC address mappings for devices on the local network. Helps troubleshoot layer 2 connectivity issues.

Common options:

- arp -a - Displays ARP cache
- arp -d - Deletes an entry
- arp -s - Adds a static entry

```
C:\Users\HP>arp -a

Interface: 10.5.5.200 --- 0x11
  Internet Address      Physical Address      Type
  10.5.50.1             74-4d-28-08-5b-e4     dynamic
  10.5.255.255          ff-ff-ff-ff-ff-ff     static
  192.168.101.1         74-4d-28-08-5b-e4     dynamic
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  224.96.237.52         01-00-5e-60-ed-34     static
  226.177.68.80         01-00-5e-31-44-50     static
  227.128.12.125        01-00-5e-00-0c-7d     static
  228.103.64.137        01-00-5e-67-40-89     static
  228.156.194.137       01-00-5e-1c-c2-89     static
  229.27.178.47         01-00-5e-1b-b2-2f     static
  229.38.210.145        01-00-5e-26-d2-91     static
  229.64.221.3          01-00-5e-40-dd-03     static
  230.62.189.197        01-00-5e-3e-bd-c5     static
  230.69.131.166        01-00-5e-45-83-a6     static
  231.210.45.136        01-00-5e-52-2d-88     static
  231.210.221.243       01-00-5e-52-dd-f3     static
  233.207.140.150       01-00-5e-4f-8c-96     static
  234.188.142.71        01-00-5e-3c-8e-47     static
  235.64.148.55         01-00-5e-40-94-37     static
  235.114.194.3         01-00-5e-72-c2-03     static
  235.131.10.100        01-00-5e-03-0a-64     static
  236.187.20.226        01-00-5e-3b-14-e2     static
  236.210.6.121         01-00-5e-52-06-79     static
  236.210.120.10        01-00-5e-52-78-0a     static
  239.42.53.114         01-00-5e-2a-35-72     static
  239.78.173.218        01-00-5e-4e-ad-da     static
  239.192.152.143       01-00-5e-40-98-8f     static
  239.250.250.250       01-00-5e-7a-fa-fa     static
  239.254.127.63        01-00-5e-7e-7f-3f     static
  239.255.102.18        01-00-5e-7f-66-12     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

7. telnet

Syntax: telnet [hostname or IP address] [port]

Usage: Connects to a remote host using the Telnet protocol, useful for testing connectivity to specific ports.

Theory: Tests whether specific TCP ports are open and accepting connections, verifying service accessibility.

Note: No output due to installation issue.

8. netsh wlan

Syntax: netsh wlan show profiles

Usage: Manages wireless network profiles and connections on Windows systems.

Theory: Provides information about saved Wi-Fi networks, security settings, and connection status for troubleshooting wireless connectivity.

```
C:\Users\HP>netsh wlan show profiles

Profiles on interface Wi-Fi:

Group policy profiles (read only)
---------------------------------
    <None>

User profiles
-------------
    All User Profile     : Bonds And Bites 2G
    All User Profile     : chiyaroma_5
    All User Profile     : Computer Lab
    All User Profile     : Himalaya01
    All User Profile     : HIMALAYA
    All User Profile     : Sneha’s iPhone
    All User Profile     : Bonds and Bites 5G
    All User Profile     : TP-Link_06C4_5G
    All User Profile     : TP-Link_06C4
    All User Profile     : chiya kuti_5
    All User Profile     : Bonds and Bites
    All User Profile     : Bonds and Bites (BNB)
    All User Profile     : Chiyadahacustomer
    All User Profile     : Redmi Note 9
    All User Profile     : khom@vianet_5G
    All User Profile     : khom@vianet
    All User Profile     : Engineer's Network
    All User Profile     : TP-Link_CBD0
    All User Profile     : Bhageshwor
    All User Profile     : Engineer's Network_5g
    All User Profile     : The S_cube caffee
    All User Profile     : ARCHITECTURE
    All User Profile     : wi_useitfree
    All User Profile     : Redmi Note 11
    All User Profile     : Free_WorldLink_3rdclass_Wi-Fi
```

9. pathping

Syntax: pathping [hostname or IP address]

Usage: Combines the functionality of ping and tracert to provide information about network latency and packet loss at each hop.

Theory: Traces the route and sends multiple packets to each router over time, calculating statistics to identify which hops cause latency or packet loss.

```
C:\Users\HP>pathping www.google.com

Tracing route to www.google.com [216.58.200.164]
over a maximum of 30 hops:
  0  LAPTOP-D4UFQDGJ [10.5.5.200]
  1  10.5.50.1
  2  172.25.25.1
  3  1.220.166.202.ether.static.wlink.com.np [202.166.220.1]
  4  be-17.3005-140.40.gwc-dhub-core-01.wlink.com.np [202.79.40.140]
  5  ae-21-170.41.gwmj-lknt-01.wlink.com.np [202.79.41.170]
  6  ae-20-128.41.gwj-btwl-core-01.wlink.com.np [202.79.41.128]
  7  ae52-ipt-bhwa-01.wlink.com.np [72.9.128.67]
  8     *        *        *
Computing statistics for 175 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           LAPTOP-D4UFQDGJ [10.5.5.200]
                                 3/ 100 =  3%   |
  1   35ms     3/ 100 =  3%     0/ 100 =  0%  10.5.50.1
                                 0/ 100 =  0%   |
  2   39ms     3/ 100 =  3%     0/ 100 =  0%  172.25.25.1
                                 0/ 100 =  0%   |
  3   37ms     6/ 100 =  6%     3/ 100 =  3%  1.220.166.202.ether.static.wlink.com.np [202.166.220.1]
                                 0/ 100 =  0%   |
  4   39ms     4/ 100 =  4%     1/ 100 =  1%  be-17.3005-140.40.gwc-dhub-core-01.wlink.com.np [202.79.40.140]
                                 0/ 100 =  0%   |
  5   34ms     5/ 100 =  5%     2/ 100 =  2%  ae-21-170.41.gwmj-lknt-01.wlink.com.np [202.79.41.170]
                                 0/ 100 =  0%   |
  6   41ms     3/ 100 =  3%     0/ 100 =  0%  ae-20-128.41.gwj-btwl-core-01.wlink.com.np [202.79.41.128]
                                 1/ 100 =  1%   |
  7   39ms     4/ 100 =  4%     0/ 100 =  0%  ae52-ipt-bhwa-01.wlink.com.np [72.9.128.67]

Trace complete.
```

10. route print

Syntax: route print

Usage: Displays the current IP routing table on the local machine.

Theory: Shows how packets are forwarded to different networks. Each entry specifies destination, gateway, interface, and metric.

```
C:\Users\HP>route print
===========================================================================
Interface List
  3...68 54 5a ad ef 4f ......Microsoft Wi-Fi Direct Virtual Adapter
 13...6a 54 5a ad ef 4e ......Microsoft Wi-Fi Direct Virtual Adapter #2
 17...68 54 5a ad ef 4e ......Intel(R) Wi-Fi 6 AX200 160MHz
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0        10.5.50.1       10.5.5.200     55
         10.5.0.0      255.255.0.0         On-link        10.5.5.200    311
       10.5.5.200  255.255.255.255         On-link        10.5.5.200    311
     10.5.255.255  255.255.255.255         On-link        10.5.5.200    311
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link        10.5.5.200    311
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link        10.5.5.200    311
===========================================================================
Persistent Routes:
  None

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
  1    331 ::1/128                  On-link
 17    311 fe80::/64                On-link
 17    311 fe80::b25b:16d9:7c03:357/128
                                    On-link
  1    331 ff00::/8                 On-link
 17    311 ff00::/8                 On-link
===========================================================================
Persistent Routes:
  None
```

11. getmac

Syntax: getmac

Usage: Displays the MAC addresses for network adapters on the local machine.

Theory: MAC addresses are unique hardware identifiers for network interface cards. Useful for identifying adapters and troubleshooting MAC filtering issues.

```
C:\Users\HP>getmac

Physical Address    Transport Name
=================== ==========================================================
68-54-5A-AD-EF-4E   \Device\Tcpip_{F72C3FB3-01F1-4833-946C-83BA1DBBEAEA}
```

12. nbtstat

Syntax: nbtstat -a [hostname]

Usage: Displays NetBIOS over TCP/IP statistics, including the NetBIOS name table of a remote computer.

Theory: NetBIOS is a legacy protocol used for Windows file sharing and networking. Helps troubleshoot Windows networking issues.

Common options:

- nbtstat -a [hostname] - NetBIOS name table by hostname
- nbtstat -A [IP] - NetBIOS name table by IP
- nbtstat -c - NetBIOS name cache

```
C:\Users\HP>nbtstat -n

Wi-Fi:
Node IpAddress: [10.5.5.200] Scope Id: []

                NetBIOS Local Name Table

       Name               Type         Status
    ---------------------------------------------
    LAPTOP-D4UFQDGJ<20>  UNIQUE      Registered
    WORKGROUP      <00>  GROUP       Registered
    LAPTOP-D4UFQDGJ<00>  UNIQUE      Registered

Local Area Connection* 1:
Node IpAddress: [0.0.0.0] Scope Id: []

    No names in cache

Local Area Connection* 2:
Node IpAddress: [0.0.0.0] Scope Id: []

    No names in cache
```

13. whois

Syntax: whois [domain name]

Usage: Retrieves registration information about a domain name from the WHOIS database.

Theory: Provides domain registration details including registrar, registration dates, and nameservers for verifying domain ownership.

Note: Some outputs were not displayed due to recognition error.

```
C:\Users\HP>whois
'whois' is not recognized as an internal or external command,
operable program or batch file.
```

Conclusion

This lab offered a practical experience on the use of different network commands that are critical in diagnosing and troubleshooting networking problems. We got a practical experience on how to ensure network configuration, test connectivity, trace network paths, monitor active connections, and solve DNS queries, through the practical work of these tools.